Week 4 · Lab 2

Version C

Graded Practice:

Operators and `if/else`

10 problems from LeetCode · 10 points each · checked by the TA at your desk

Week: 4 — second session (graded)
Duration: 2 hours 15 minutes
Format: individual problem solving in CLion
Covers: Lecture 3 — operators, short-circuit, `if/else if`, nested `if`, output formatting (`<iomanip>`)

Name:

Group:

Date:

How this lab works

1. Solve the problems in CLion, one `.cpp` file per problem. Every program reads its input with `cin` and prints its answer with `cout`.
2. **When a problem works, raise your hand.** The TA comes to your desk, reads your code, runs a few inputs and writes your points in the box under the problem. Nothing is uploaded.
3. There are **10 problems, 10 points each** (100 in total): problems 1–4 are **easy**, 5–7 **medium** and 8–10 **hard**. Solve them in any order; a problem with no attempt earns 0 points.
4. Under each title, **Practises:** says which part of Lecture 3 the problem trains.
5. You may ask the TA for help — this is practice, not an exam. The TA may ask you a short question about your code; be ready to explain *why* it works.

All problems are from `leetcode.com`, restated for a program that reads from the keyboard (some are restricted to a fixed number of inputs, because loops and arrays come later). On LeetCode itself the answer is written as a *function* inside `class Solution`; you will learn functions in Week 6.

Quick reference (Lecture 3)

Arithmetic `17 / 5` is 3 and `17 % 5` is 2 (`int / int` throws the fraction away); `-7 / 2` is `-3`. `n % 10` is the last digit, `n / 10 % 10` the tens digit. Convert *before* dividing: `static_cast<double>(a) / b`. Compound forms: `x += 5; x %= 3;` `x++` / `++x` add 1.

Precedence, highest first: `()` · `!` · `* / %` · `+ -` · `< <= > >=` · `== !=` · `&&` · `||` · `=`. Inside `cout`, put a comparison in parentheses: `cout << (a < b);`

Conditions A range is two tests: `3 < x && x < 5` (`3 < x < 5` compiles but is always true). `&&` stops at the first false, `||` at the first true, so put the guard on the left:

```
if (c != 0 && a / c > 1) { ... }   // never divides by 0
```

Give long conditions a name, and remember that in arithmetic `true` counts as 1 and `false` as 0:

```
bool leap = (y % 4 == 0 && y % 100 != 0) || y % 400 == 0;
int odd = (a % 2 == 1) + (b % 2 == 1);   // 0, 1 or 2
```

Reading formatted input `cin >> c` for a `char` skips spaces and reads one character, so separators go into a spare `char`:

```
int y, m, d;  char dash;
cin >> y >> dash >> m >> dash >> d;     // 2019-02-10
```

`else if` chains The *first* true condition wins and the rest are skipped — test the strictest case first.

```
if (avg >= 90) {
    cout << "Excellent";
} else if (avg >= 75) {      // here avg < 90 already
    cout << "Good";
} else {
    cout << "Needs work";
}
```

Nested `if` An `else` binds to the *nearest* unmatched `if`. Always use braces.

Maximum, minimum, absolute value

```
int mx = a;
if (b > mx) { mx = b; }
if (c > mx) { mx = c; }      // mx = max(a, b, c)
int d = a - b;
if (d < 0) { d = -d; }       // d = |a - b|
```

Formatting output `#include <iomanip>`

```
cout << fixed << setprecision(2) << 2.5;   // 2.50
cout << setfill('0') << setw(3) << 7;      // 007
```

`fixed`, `setprecision` and `setfill` stay in force; `setw` applies to the *next item only*. Without `fixed`, `setprecision(2)` means 2 *significant* digits.

Classic bugs `if (x = 5)` (use `==`); `if (x > 0);` (stray `;`); `x == 6 || 28` (always true — write `x == 6 || x == 28`); `int` overflow (use `long long`).

EASY

Problems 1–4 · 40 points

Short programs: one idea each. Test every branch, not just the examples.

1 Count Distinct Numbers on Board 10 pts

LeetCode 2549 · `leetcode.com/problems/count-distinct-numbers-on-board`

Practises: `if/else` — and an edge case.

A positive integer n ($1 \le n \le 100$) is written on a board. Every day, for every number x on the board, you also write every i with $1 \le i \le n$ and $x \,\%\, i = 1$. Numbers are never erased. After 10^9 days, how many *different* numbers are on the board?

Input	Output	Why
5	4	5%2 = 1 and 5%4 = 1 add 2 and 4; then 4%3 = 1 adds 3. Board: 2, 3, 4, 5.
3	2	3%2 = 1 adds 2; nothing else. Board: 2, 3.

Hint: which number does n always add? And which number does that *one add?*

TA: ____ / 10 pts initials ____

2 Neither Minimum nor Maximum (3 numbers) 10 pts

LeetCode 2733 · `leetcode.com/problems/neither-minimum-nor-maximum`

Practises: `&&` and `||` in an `else if` chain.

Read three **different** integers (each from 1 to 100) and print the one that is neither the smallest nor the largest. (LeetCode gives an array of any length; here it is always three numbers.)

Input	Output	Why
3 2 1	2	
5 9 7	7	

Hint: `a` *is the middle one if it is bigger than one of the others and smaller than the other. Two ways round, so* `||`.

TA: ____ / 10 pts initials ____

3 Perfect Number 10 pts

LeetCode 507 · `leetcode.com/problems/perfect-number`

Practises: one condition with `||`.

A *perfect number* equals the sum of its positive divisors other than itself: 28 = 1 + 2 + 4 + 7 + 14. Read `num` ($1 \le$ `num` $\le 10^8$) and print `true` if it is perfect, otherwise `false`.

Input	Output	Why
28	`true`	
7	`false`	$1 \neq 7$.

Hint: finding the divisors needs a loop — but up to 10^8 *there are only five perfect numbers: 6, 28, 496, 8128 and 33550336. (This is an accepted LeetCode solution.)*

TA: ____ / 10 pts initials ____

4 Add Digits 10 pts

LeetCode 258 · `leetcode.com/problems/add-digits`

Practises: `if/else` and `%`.

Read `num` ($0 \le$ `num` $\le 2^{31} - 1$). Add up its digits; if the result has more than one digit, add up *its* digits; repeat until a single digit is left, and print it. Do it without a loop.

Input	Output	Why
`38`	`2`	3 + 8 = 11, then 1 + 1 = 2.
`0`	`0`	

Hint: write down the answer for `num` $= 1, 2, \ldots, 20$ *and look for the pattern. Which operator repeats every 9 numbers? Treat 0 separately.*

TA: _____ / 10 pts initials _____

MEDIUM

Problems 5–7 · 30 points

A few cases to get right. Test the boundaries.

5 Valid Boomerang 10 pts

LeetCode 1037 · `leetcode.com/problems/valid-boomerang`

Practises: one condition with `&&` / `||`.

Read three points (x_1, y_1), (x_2, y_2), (x_3, y_3), each coordinate from 0 to 100. They form a *boomerang* if all three are different and they do **not** lie on one straight line. Print `true` or `false`.

Input	Output	Why
`1 1 2 3 3 2`	`true`	
`1 1 2 2 3 3`	`false`	All on the line $y = x$.

Hint: the points are on one line when the slopes from point 1 are equal: $\frac{y_2-y_1}{x_2-x_1} = \frac{y_3-y_1}{x_3-x_1}$. *Multiply out instead of dividing — why?*

TA: ____ / 10 pts initials ____

6 Number of Valid Clock Times 10 pts

LeetCode 2437 · `leetcode.com/problems/number-of-valid-clock-times`

Practises: nested `if` on `chars`.

A time is written `hh:mm`; valid times are `00:00` to `23:59`. Some digits are unknown and shown as `?`. Read such a time and print how many valid times it could be.

Input	Output	Why
`?5:00`	`2`	`05:00` or `15:00` (`25:00` is not valid).
`0?:0?`	`100`	10 choices for each `?`.
`??:??`	`1440`	$24 \cdot 60$.

Hint: read five `chars`. *Count the possible hours and the possible minutes separately, then multiply. The hours need four cases: both digits unknown, only the first, only the second, none.*

TA: ____ / 10 pts initials ____

7 Integer to Roman (1 to 99) 10 pts

LeetCode 12 · `leetcode.com/problems/integer-to-roman`

Practises: `else if` and nested `if`, on the tens and the ones digit.

Roman numerals use `I` = 1, `V` = 5, `X` = 10, `L` = 50, with the subtractive forms `IV` = 4, `IX` = 9, `XL` = 40 and `XC` = 90. Read `num` ($1 \le$ `num` ≤ 99) and print it as a Roman numeral. (LeetCode goes up to 3999; here we stop at 99.)

Input	Output	Why
`58`	`LVIII`	50 + 5 + 3.
`3`	`III`	
`94`	`XCIV`	90 + 4.

Hint: the tens digit and the ones digit can be written separately, with the same pattern: 9 and 4 are special; otherwise print the 5-symbol if the digit is 5 or more, then 0–3 one-symbols. Print `endl` *only at the very end.*

TA: ____ / 10 pts initials ____

HARD

Problems 8–10 · 30 points

More cases and more care. Read the hint before you start.

8 Count Days Spent Together 10 pts

LeetCode 2409 · `leetcode.com/problems/count-days-spent-together`

Practises: conditions with `&&` / `||`, and `else if` chains.

Alice and Bob are both visiting Rome in the same year, which is **not** a leap year. Read `arriveAlice leaveAlice arriveBob leaveBob`, each written `MM-DD` (arrival never after departure). Print how many days they are both in Rome, counting arrival and departure days.

Input	Output	Why
`08-15 08-18 08-16 08-19`	`3`	16, 17 and 18 August.
`10-01 10-31 11-01 12-31`	`0`	

Hint: `cin >> m >> dash >> d;` *reads* `08-15`*. They are together from the later arrival to the earlier departure. Turn those two dates into day numbers with an* `else if` *chain on the month: days before January 0, before February 31, before March 59, . . . , before December 334.*

TA: ____ / 10 pts initials _____

9 Best Poker Hand 10 pts

LeetCode 2347 · `leetcode.com/problems/best-poker-hand`

Practises: a `bool` for the flush, a maximum, then an `else if` chain.

Read five card ranks (each 1 to 13), then the five suits (each `a`, `b`, `c` or `d`); no two cards are identical. Print the best hand: `Flush` (all five suits the same), else `Three of a Kind` (at least three cards of one rank), else `Pair` (two cards of one rank), else `High Card`.

Input	Output	Why
`13 2 3 1 9 a a a a a`	`Flush`	
`4 4 2 4 4 d a a b c`	`Three of a Kind`	Four 4s still count as three of a kind.
`10 10 2 12 9 a b c a d`	`Pair`	

Hint: in arithmetic `true` *counts as 1, so* `1 + (r1 == r2) + (r1 == r3) + (r1 == r4) + (r1 == r5)` *counts the cards that share card 1's rank.*

TA: ____ / 10 pts initials _____

10 Thousand Separator 10 pts

LeetCode 1556 · `leetcode.com/problems/thousand-separator`

Practises: an `else if` chain, and `setfill`/`setw` from `<iomanip>` for the zeros.

Read n ($0 \le n \le 2^{31} - 1$) and print it with a dot `.` between every group of three digits, counting from the right.

Input	Output	Why
`987`	`987`	
`1234`	`1.234`	
`2147483647`	`2.147.483.647`	

Hint: `n % 1000` *is the last group,* `n / 1000 % 1000` *the next one, and so on. Every group except the first must be printed with exactly three digits: 7 as* `007`*, 45 as* `045` *—* `setfill('0') << setw(3)` *does it, but remember that* `setw` *applies to the next item only.*

TA: ____ / 10 pts initials _____

Score sheet — Version C

	Problem	Level	Points	Earned	TA
1	Count Distinct Numbers on Board (LC 2549)	Easy	10		
2	Neither Minimum nor Maximum (3 numbers) (LC 2733)	Easy	10		
3	Perfect Number (LC 507)	Easy	10		
4	Add Digits (LC 258)	Easy	10		
5	Valid Boomerang (LC 1037)	Medium	10		
6	Number of Valid Clock Times (LC 2437)	Medium	10		
7	Integer to Roman (1 to 99) (LC 12)	Medium	10		
8	Count Days Spent Together (LC 2409)	Hard	10		
9	Best Poker Hand (LC 2347)	Hard	10		
10	Thousand Separator (LC 1556)	Hard	10		
	Total		**100**		

After the lab

Every problem here is on LeetCode under the number shown. Some were cut down to a fixed number of inputs; with the loops of Week 5 you can solve the full versions. Once functions are covered (Week 6), rewrite your solutions as a LeetCode `class Solution` and submit them.

End of Week 4 · Lab 2 · Version C